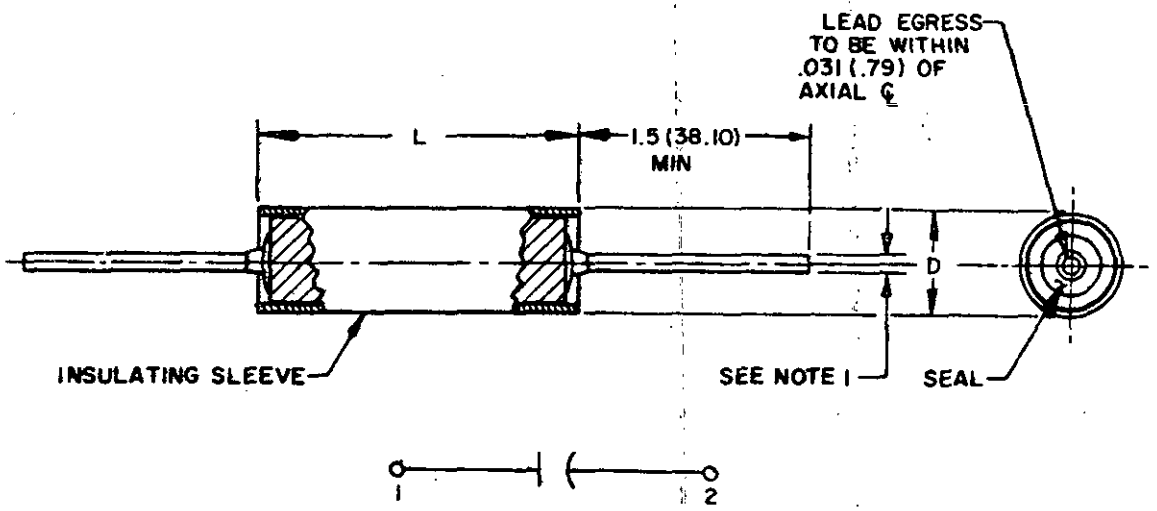
MILITARY SPECIFICATION SHEET

Ⓐ CAPACITORS, FIXED, METALLIZED, PAPER-PLASTIC FILM DIELECTRIC, DIRECT CURRENT, (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLE CHR29 (INSULATED)

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39022.



CIRCUIT DIAGRAM

NOTES:
1. Number 22 AWG wire for case diameters of .360 (9.14 mm) and under. Number 20 AWG wire for case diameters over .360 (9.14 mm).
2. See table I for additional dimensions.
3. Dimensions are in inches.
4. Metric equivalents (to the nearest .01 mm) are given for general information only, and are based upon 1 inch = 25.4 mm.
5. Insulating sleeve shall extend beyond the capacitor body but shall not exceed .031 (.79 mm) inch on either end. Insulating sleeve thickness shall not exceed .016 (.41 mm) inch.
6. Metric equivalents are in parentheses.

Ⓐ denotes changes.

FSC 5910

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(A) TABLE I. Electrical characteristics, dimensions and dash numbers.

DC rated voltage	Capacitance value	Dimension, maximum (inches) 1/ 2/		Dash number Failure rate level for (%/1, 000 hr)			
		L	D	M (1.0)	P (0.1)	R (0.01)	S (0.001)
Volts	µF						
200	0.0047	.907	.223	1001	1051	1101	1151
200	0.0056	.907	.223	1002	1052	1102	1152
200	0.0068	.907	.223	1003	1053	1103	1153
200	0.0082	.907	.223	1004	1054	1104	1154
200	0.010	.907	.223	1005	1055	1105	1155
200	0.012	.907	.243	1006	1056	1106	1156
200	0.015	.907	.243	1007	1057	1107	1157
200	0.018	.907	.283	1008	1058	1108	1158
200	0.022	.907	.283	1009	1059	1109	1159
200	0.027	.907	.283	1010	1060	1110	1160
200	0.033	.907	.283	1011	1061	1111	1161
200	0.039	1.032	.360	1012	1062	1112	1162
200	0.047	1.032	.360	1013	1063	1113	1163
200	0.056	1.032	.360	1014	1064	1114	1164
200	0.068	1.032	.360	1015	1065	1115	1165
200	0.082	1.032	.360	1016	1066	1116	1166
200	0.10	1.032	.360	1017	1067	1117	1167
200	0.12	1.282	.360	1018	1068	1118	1168
200	0.15	1.282	.360	1019	1069	1119	1169
200	0.18	1.032	.448	1020	1070	1120	1170
200	0.22	1.032	.448	1021	1071	1121	1171
200	0.27	1.282	.448	1022	1072	1122	1172
200	0.33	1.282	.448	1023	1073	1123	1173
200	0.39	1.282	.548	1024	1074	1124	1174
200	0.47	1.282	.548	1025	1075	1125	1175
200	0.56	1.282	.548	1026	1076	1126	1176
200	0.68	1.282	.548	1027	1077	1127	1177
200	0.82	1.282	.610	1028	1078	1128	1178
200	1.0	1.282	.610	1029	1079	1129	1179
200	1.5	1.782	.610	1030	1080	1130	1180
200	2.0	1.782	.718	1031	1081	1131	1181
200	3.0	2.032	.718	1032	1082	1132	1182
200	4.0	2.032	.798	1033	1083	1133	1183
200	5.0	2.032	1.048	1034	1084	1134	1184
200	6.0	2.032	1.048	1035	1085	1135	1185
200	8.0	2.032	1.048	1036	1086	1136	1186
200	10.0	2.532	1.048	1037	1087	1137	1187
200	12.0	2.782	1.048	1038	1088	1138	1188
400	0.68	2.032	.718	1039	1089	1139	1189

1/ Dimensions include insulating sleeving.

2/ Metric equivalents see table II.

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Ⓐ TABLE II. Metric equivalents of decimal inches. 1/

Inches	MM	Inches	MM	Inches	MM
.223	5.66	.610	15.49	1.282	32.56
.243	6.17	.718	18.24	1.782	45.26
.283	7.19	.798	20.27	2.032	51.61
.360	9.14	.907	23.04	2.532	64.31
.448	11.38	1.032	26.21	2.782	70.66
.548	13.92	1.048	26.62		

1/ Metric equivalents (to the nearest .01 millimeter) are given for general information only, and are based upon 1 inch = 25.4 mm.

REQUIREMENTS:

Dimensions and configuration: See figure and table I.

Case material: Nonmagnetic.

Capacitance value: See table I.

Capacitance tolerance: ±5 percent.

DC Rated voltage: See table I.

Ⓐ Dielectric material: Normally paper-polyethylene-terephthalate.

Operating temperature range: -55° to +125° C.

Failure rate level: M (1.0%), P (0.1%), R (0.01%), and S (0.001%).

Temperature cycling: Method 102 of MIL-STD-202, condition D.

Seal: Method 112 of MIL-STD-202, condition A, or (for liquid-filled units only) 1 hour at +125° +4, -0° C with no evidence of leakage.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202.
Terminal to terminal - 200 percent of rated voltage.
Terminal to case - 200 percent of rated voltage.

NOTE: For 100-percent inspection:
250 percent for not less than 5 seconds, or
200 percent for not less than 15 seconds.

Insulation resistance (IR): Method 302 of MIL-STD-202.
Terminal to terminal - See table III.
Terminals to case - 10,000 megohms, minimum.

Dissipation factor (DF): 1 percent.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100,000 feet).
125 percent of rated voltage applied.

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Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20G). 50 percent of rated voltage applied.

TABLE III. Insulation resistance.

	At dc voltage of	
	200V	400V
In megohms:		
At +25° ±3°C (need not exceed)	12,000	12,000
At +125° $^{+4°}_{-0°}$C (need not exceed)	150	600
In megohms x microfarads (minimum):		
At +25° ±3°C	2,000	2,000
At +125° $^{+4°}_{-0°}$C	10	40

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours). Salt solution - 5 percent.

Immersion: Method 104 of MIL-STD-202, condition C.
DWV:
Insulating sleeves - Greater than 4,000 volts dc.
Terminal to terminal - 200 percent of rated voltage.
Terminals to case - 200 percent of rated voltage.
IR:
Insulating sleeves - 100 megohms, minimum.
Terminal to terminal - Not less than 50 percent of value specified in table III.
Terminals to case - 5,000 megohms, minimum.
Cap. - Within ±10 percent of initial value.
DF - Not more than 1.1 percent.

(A) Shock (specified pulse): Method 213 of MIL-STD-202, condition I. 50 percent of rated voltage applied.

Moisture resistance: Method 106 of MIL-STD-202.
DWV, IR, Cap., and DF - Same as for immersion.

Terminal strength: Method 211 of MIL-STD-202, conditions A and D.
Condition A - Applied force 4-1/2 pounds.
Condition D - 3 rotations of 360 degrees.

Low temperature and cap. change with temperature:
Low temperature: -55° +0°, -5°C for 48 hours with rated voltage applied.
Cap. change with temperature:
At -55°+0°, -5°C - -10 percent, maximum.
At +125°+4°, -0°C - +20 percent, maximum.

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Fault count: In accordance with MIL-C-39022.
At +125°+4, -0°C monitored for momentary breakdowns and periods of low IR.
Qualification - 2,000 +8, -0 hours; quality conformance (group C) 250 +8, -0 hours.
Breakdowns - Shall not exceed 6C or 1, whichever is greater (where C is the nominal capacitance in μF of the sample). Total counts for any single capacitor in a 24-hour period shall not exceed 4C or 1, whichever is greater.
Period of low IR - Not to exceed 2 minutes.
IR:
Terminal to terminal - Not less than 33.3 percent of initial value.
Terminals to case - 5,000 megohms, minimum.
Cap. - Within ±10 percent of initial measured value.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B (260 ±5°C for 10 ±1 seconds).
IR - See table III.
Cap. - Within ±5 percent of initial value.
DF - Not more than 1.0 percent.

Life: Method 108 of MIL-STD-202.
Accelerated conditions: 140 percent of rated voltage for 2,000 +72, -0 hours.
Rated conditions: 100 percent of rated voltage for 10,000 +96, -10 hours.
DF (at +125°+4°, -0°C) between 24 and 48 hours of test - 2 percent, maximum.
During last 48 hours of test - 2.5 percent, maximum.
IR:
Insulating sleeves - 100 megohms, minimum.
Terminal to terminal - Not less than 33.3 percent of initial value.
Terminals to case - 5,000 megohms, minimum.
Cap. - Within ±10 percent of initial value.
DF (at +25° ±3°C) after life - 1.1 percent.

(A) Marking: In accordance with MIL-C-39022.

Part number: M39022/04- (dash number from table I)

(A) Cross reference data from type designations of MIL-C-39022/4(USAF) to part numbers in MIL-C-39022/4A(USAF) are not furnished since a qualified source did not exist at the time of this revision.

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